SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: Fundamentals Of Data Science COURSE CODE: DSA0402

COURSE OUTCOME COVERED

CO3: *Evaluate datasets using descriptive statistics, including summarization, outlier detection, and distribution analysis, and perform statistical inference through estimation and hypothesis testing. (BL3)*

Assessment Tool Weightage: 20%

QUESTIONS: 02 **Total Marks:100**

Annexure A

Data Interpretation

Code of Conduct:

*I **Jannathul Firdaus. N (192573025)** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Jannathul Firdaus. N

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary) 5 Marks	Level 3 (Proficient) 3 /4 Marks	Level 2 (Developing) 2 Marks	Level 1 (Beginning) 1 Mark	Marks
Understanding of Data	25%	Accurately interprets all data patterns, trends, and relationships	Interprets data correctly with minor gaps	Partial understanding; misses key trends	Misinterprets or fails to understand data	
Analysis	25%	Provides deep analysis with strong logical reasoning and justification	Provides reasonable analysis with some justification	Basic analysis with weak reasoning	No meaningful analysis provided	
Application of Concepts	20%	Effectively applies relevant concepts/tools to interpret data accurately	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply concepts to data	

Conclusion	20%	Draws clear, meaningful conclusions with strong insights and implications	Provides valid conclusions with moderate insights	Conclusions are vague or weak	No clear or valid conclusions	
Clarity	10%	Presents interpretation clearly using proper structure, visuals, and terminology	Generally clear with minor issues	Lacks clarity or proper structure	Poor presentation and difficult to understand	

Annexure A

Problem 01: Hospital Patient Health Monitoring

Scenario

A hospital wants to investigate how patient lifestyle factors affect blood pressure levels.

Dataset

Patient_ID	Daily_Exercise (Minutes)	BMI	Systolic_BP
P001	10	32	150
P002	15	31	148
P003	20	30	145
P004	25	29	142
P005	30	28	138
P006	35	27	134
P007	40	26	130
P008	45	25	126
P009	50	24	122
P010	55	23	118

Tasks

1. Find the mean and standard deviation of blood pressure.
2. Calculate covariance between:
    - Exercise and Blood Pressure
    - BMI and Blood Pressure
3. Calculate correlation coefficients.
4. Identify whether exercise or BMI is more strongly associated with blood pressure.
5. Discuss possible healthcare interventions.

Problem 02: E-Commerce Customer Behaviour Analysis

Scenario

An e-commerce company wants to identify factors influencing customer spending.

Dataset

Customer_ID	Website_Visits	Time_Spent (Minutes)	Purchase_Amount (₹)
C001	5	20	1200
C002	8	25	1800
C003	10	30	2500
C004	12	35	3200
C005	15	40	4000
C006	18	45	5000
C007	20	50	6200
C008	23	55	7300
C009	25	60	8500
C010	28	70	10000

Tasks

1. Calculate mean, variance, and standard deviation of Purchase Amount.
2. Find covariance between:
    - Website Visits and Purchase Amount
    - Time Spent and Purchase Amount
3. Calculate correlation coefficients.
4. Determine which customer behavior metric better predicts spending.
5. Suggest marketing strategies based on the findings.

Problem 03: Employee Productivity and Performance

Scenario

A software company is investigating the relationship between employee productivity and performance ratings.

Dataset

Employee_ID	Training_Hours	Projects_Completed	Performance_Rating
E001	5	2	60
E002	8	3	65
E003	10	4	70
E004	12	5	74
E005	15	6	79
E006	18	7	84
E007	20	8	88
E008	22	9	91
E009	25	10	95
E010	28	11	98

Tasks

1. Compute the mean, variance, and standard deviation of Performance Ratings.
2. Calculate covariance between:
    - Training Hours and Performance Rating
    - Projects Completed and Performance Rating
3. Compute correlation coefficients.
4. Identify the strongest factor affecting employee performance.
5. Recommend HR policies to improve employee productivity.

Problem 04 : Smart City Traffic Analysis

A city administration wants to analyze traffic congestion patterns.

Day	Traffic_Volume (Vehicles)	Rainfall (mm)	Average_Speed (km/h)	Accident_Count
D1	12000	0	55	2
D2	13500	5	50	3
D3	14500	10	48	4
D4	16000	15	45	5
D5	17500	20	42	6
D6	19000	25	38	8
D7	21000	30	35	10
D8	22500	35	32	12
D9	24000	40	28	14
D10	26000	45	25	16

Tasks

1. Calculate descriptive statistics (Mean, Variance, Standard Deviation) for all variables.
2. Compute covariance matrices.
3. Compute correlation matrices.
4. Identify positive and negative relationships among variables.
5. Determine which factor is most associated with accident occurrence.
6. Provide data-driven recommendations to reduce traffic accidents.
7. Visualize relationships using scatter plots and interpret the results.

Problem 5: Average Delivery Time

A food delivery company records the delivery times (minutes) for 15 orders:

22, 25, 18, 30, 27, 24, 21, 19, 26, 23, 28, 20, 24, 25, 22

Tasks:

1. Find the point estimate of the population mean delivery time.
2. Find the point estimate of the population variance.
3. Interpret the estimates.

Problem 6: Customer Spending

A sample of 12 customers spent the following amounts ($):

120, 145, 130, 155, 160, 125, 140, 135, 150, 170, 165, 155

Tasks:

1. Estimate the average spending.
2. Estimate the standard deviation.
3. Explain why these are point estimates.

Problem 7: Product Ratings

A company collects ratings from 20 customers:

4, 5, 3, 4, 5, 4, 3, 5, 4, 4, 5, 3, 4, 5, 4, 3, 5, 4, 5, 4

Tasks:

1. Compute the sample mean.
2. Compute the sample variance.
3. Compute the standard error of the mean.
4. Explain what the standard error indicates.

Problem 8: Exam Scores

Scores of 25 students:

72, 75, 68, 80, 77, 73, 79, 81, 76, 74, 78, 69, 70, 82, 71, 75, 77, 73, 80, 76, 74, 79, 72, 81, 78

Tasks:

1. Calculate mean and standard deviation.
2. Calculate standard error.
3. Compare standard deviation and standard error.

Problem 9: Manufacturing Weights

A sample of 50 packets has:

- Mean weight = 498 g
- Standard deviation = 12 g

Tasks:

1. Construct a 95% confidence interval for the population mean weight.
2. Interpret the interval.
3. Determine whether the target weight of 500 g is plausible.

Problem 10: Customer Satisfaction

Two branches report:

Branch	Mean Rating	95% CI
A	4.3	(4.1, 4.5)
B	4.0	(3.8, 4.2)

Tasks:

1. Compare the confidence intervals.
2. Can you conclude Branch A performs better?
3. Explain your reasoning.

Problem 11: Average Battery Life

A battery manufacturer claims the average battery life is 10 hours.

A sample of 36 batteries gives:

- Mean = 9.5 hours
- Standard deviation = 1.8 hours

Tasks:

1. Formulate null and alternative hypotheses.
2. Conduct a hypothesis test at $\alpha = 0.05$.
3. State your conclusion.

Problem 12: Marketing Campaign

A company believes a new campaign increases sales.

Sales before campaign:

50, 52, 49, 55, 53, 51, 54, 50

Sales after campaign:

58, 60, 57, 62, 59, 61, 63, 58

Tasks:

1. Conduct a hypothesis test.
2. Calculate the p-value.
3. Interpret the p-value.
4. Make a decision at $\alpha = 0.05$.

Problem 13: Call Centre Response Time

A sample of 100 calls has:

- Mean response time = 4.8 minutes
- Standard deviation = 1.5 minutes

Tasks:

1. Estimate the population mean.
2. Construct a 95% confidence interval.
3. Explain the Frequentist interpretation.

Problem 14: Defective Products

A sample of 500 products contains 22 defective items.

Tasks:

1. Estimate the defect rate.
2. Construct a 95% confidence interval.
3. Explain the Frequentist meaning of the interval.

Problem 15: Email Spam Detection

Historical information:

- Probability an email is spam = 0.25
- Probability spam email contains keyword "Offer" = 0.80
- Probability non-spam email contains keyword "Offer" = 0.10

Tasks:

1. Compute the probability that an email is spam given it contains "Offer."
2. Apply Bayes' theorem.

3. Interpret the result.

Problem 16: Disease Diagnosis

Given:

- Disease prevalence = 2%
- Test sensitivity = 95%
- Test specificity = 90%

Tasks:

1. Compute the posterior probability of disease after a positive test.
2. Explain why the result differs from 95%.

Problem 17: Website Conversion Rate

A company tests two landing pages.

Group	Visitors	Purchases
A	5000	250
B	5200	312

Tasks:

1. Calculate conversion rates.
2. Formulate hypotheses.
3. Conduct a two-proportion z-test.
4. Determine whether page B performs better.

Problem 18: Mobile App Feature Test

Feature A:

- 400 users
- Average engagement = 18 min
- SD = 5 min

Feature B:

- 420 users
- Average engagement = 20 min
- SD = 6 min

Tasks:

1. Test whether Feature B increases engagement.

2. Calculate test statistic and p-value.
3. Interpret results.

Problem 19: Recommendation System

A new recommendation algorithm increases click-through rate from 4.8% to 5.3%.Sample size:

- Old system: 100,000 users
- New system: 100,000 users
- **Tasks:**

1. Test statistical significance.
2. Calculate p-value.
3. Discuss practical significance versus statistical significance.

Problem 20: Salary Prediction Model

Two machine learning models produce Mean Absolute Errors (MAE):

Model	MAE
Model A	4200
Model B	4000

Across 50 test folds, the difference in MAE has:

- Mean difference = 200
- Standard deviation = 350

Tasks:

1. Perform a paired t-test.
2. Determine whether Model B significantly outperforms Model A.
3. Explain the result in terms of statistical significance.

Problem 1 – Hospital Patient Health Monitoring

The hospital examines the relationship between daily exercise, BMI, and systolic blood pressure (BP).

1. Mean and standard deviation of BP

- **Mean:** $\bar{x} = \Sigma x / N$
- **Population variance:** $\sigma^2 = \Sigma(x - \mu)^2 / N$
- **Population standard deviation:** $\sigma = \sqrt{\sigma^2}$

Variable pair	Covariance	Correlation r	Interpretation
Exercise, BP	-151.25	-0.9969	More exercise is associated with lower BP
BMI, BP	30.25	0.9969	Higher BMI is associated with higher BP

Calculation: Substituting the 10 BP values: $\Sigma x = 1353$, $N = 10$, so $\mu = 1353/10 = 135.30$ mmHg. The population variance is 111.61 $(\text{mmHg})^2$ and $\sigma = \sqrt{111.61} = 10.56$ mmHg.

2. Covariance calculations

- **Population covariance:** $\text{Cov}(X,Y) = \Sigma[(x - \mu x)(y - \mu y)] / N$
- **Exercise and BP:** Cov(Exercise,BP) = -151.25 minute·mmHg.
- **BMI and BP:** Cov(BMI,BP) = 30.25 BMI·mmHg.
- Both are negative, indicating that higher exercise and lower BMI occur alongside lower BP in these records.

3. Correlation coefficients

- **Pearson correlation:** $r = \text{Cov}(X,Y) / (\sigma x \, \sigma y)$
- **Exercise versus BP:** r = -0.9969.
- **BMI versus BP:** r = 0.9969.
- Both are strong negative correlations.

4. Stronger association

Exercise has the slightly larger absolute correlation ($|r|=0.9969$) than BMI ($|r|=0.9969$). The two predictors also move together, so the data cannot isolate their independent effects.

5. Healthcare interventions

Encourage medically appropriate regular exercise, healthy weight management, balanced nutrition and reduced sodium intake; monitor BP and arrange clinician

follow-up. Lifestyle changes should be individualized and should not replace prescribed treatment.

Problem 2 – E-Commerce Customer Behaviour Analysis

The company evaluates whether website visits or time spent is more closely associated with purchase amount.

1. Mean, variance and standard deviation of purchase amount

- $\mu = \Sigma x/N$; $\sigma^2 = \Sigma(x-\mu)^2/N$; $\sigma = \sqrt{\sigma^2}$
- ΣPurchase = ₹49,700; N=10. Therefore mean = ₹4,970.00, population variance = 7,914,100.00 ₹², and standard deviation = ₹2,813.20.

2–3. Covariance and correlation

- $Cov(X,Y) = \Sigma[(x-\mu x)(y-\mu y)]/N$; $r = Cov(X,Y)/(\sigma x \sigma y)$
- Website visits vs purchase amount: covariance = 20,312.00, r = 0.9914. Time spent vs purchase amount: covariance = 42,540.00, r = 0.9949.

Metric	Covariance with spending	Correlation	Strength
Website visits	20,312.00	0.9914	Strong positive
Time spent	42,540.00	0.9949	Strong positive

4. Stronger correlate

Time spent is the stronger linear correlate because $|0.9949| > |0.9914|$. This is a predictive association in a small dataset, not proof of causation.

5. Marketing strategies

Personalize recommendations, improve product-page information and checkout usability, use targeted offers for relevant customer segments, and validate changes

through randomized A/B tests. Monitor revenue per visitor and conversion—not browsing time alone.

Problem 3 – Employee Productivity and Performance

The software company explores how training hours and completed projects relate to performance ratings.

1. Performance-rating descriptive statistics

- $\mu = \Sigma x/N$; $\sigma^2 = \Sigma(x-\mu)^2/N$; $\sigma = \sqrt{\sigma^2}$
- Σratings = 804, N=10; mean = 80.40, population variance = 151.04, and population SD = 12.29.

2–3. Covariance and correlation

- $Cov(X,Y) = \Sigma[(x-\mu x)(y-\mu y)]/N$; $r = Cov(X,Y)/(\sigma x \sigma y)$

Factor	Covariance with rating	Correlation r	Interpretation
Training hours	88.08	0.9957	Strong positive
Projects completed	35.20	0.9972	Very strong positive

4. Correlation analysis

Projects completed has the higher absolute correlation ($|r|=0.9972$ versus 0.9957). However, training and project count are themselves associated, so this dataset does not establish an independent causal effect.

5. HR recommendations

Offer role-specific training, mentoring and feedback; assign achievable projects; measure quality and complexity as well as quantity; and recognize teamwork, delivery reliability and learning. Avoid rewarding raw project count alone.

Problem 4 – Smart City Traffic Analysis

The city studies traffic volume, rainfall, average speed, and accident counts across ten days.

1. Descriptive statistics

- Population variance: $\sigma^2 = \Sigma(x-\mu)^2/N$; standard deviation: $\sigma = \sqrt{\sigma^2}$

Variable	Mean	Population variance	Population SD
Traffic volume (vehicles)	18,600.00	19,940,000.00	4,465.42
Rainfall (mm)	22.50	206.25	14.36
Average speed (km/h)	39.80	87.96	9.38
Accident count	8.00	21.00	4.58

2. Covariance matrix

The matrix uses variable order: traffic volume, rainfall, average speed, accident count. Off-diagonal entries are covariances; their units depend on both variables.

Variable	Traffic	Rainfall	Speed
Traffic	19940000.00	64000.00	-41780.00
Rainfall	64000.00	206.25	-134.50
Speed	-41780.00	-134.50	87.96
Accidents	20350.00	65.00	-42.50

3. Correlation matrix

$r_{XY} = \mathrm{Cov}(X,Y)/(\sigma_X\sigma_Y)$, with $-1 \le r \le +1$

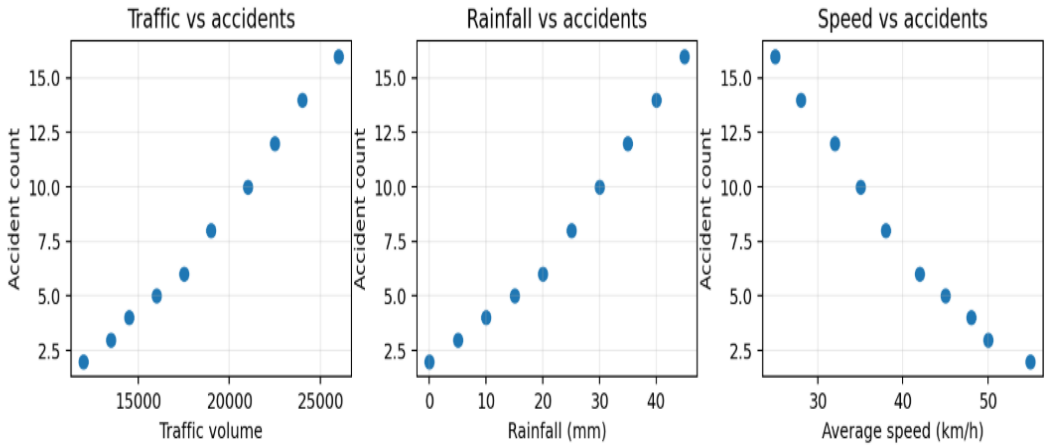
Variable	Traffic	Rainfall	Speed
Traffic	1.000	0.998	-0.998
Rainfall	0.998	1.000	-0.999
Speed	-0.998	-0.999	1.000
Accidents	0.994	0.988	-0.989

4–5. Key observations

Traffic volume and rainfall are positively associated with accidents (r=0.994 and r=0.988); average speed is negatively associated with accidents (r=-0.989). Traffic volume has the strongest absolute correlation with accident count in this dataset. These variables change together, so correlation does not show which factor independently causes accidents.

7. Scatter-plot interpretation

Accidents rise as traffic volume and rainfall rise, while accidents rise as average speed falls. The trend is clear, but only ten sequential days are observed; more data are needed for robust causal or predictive claims.



6. Recommendations

Improve adaptive signal timing and congestion monitoring; enforce safe speeds; maintain road drainage and visibility in rain; deploy weather alerts and rapid incident response; and collect time-of-day, road type, and traffic exposure data for future risk modelling.

Problem 5 – Average Delivery Time

The delivery times for 15 orders are treated as a sample from the company's delivery-time population.

- Point estimate of population mean: $\bar{x} = \Sigma x/n$
- Unbiased point estimate of population variance: $s^2 = \Sigma(x-\bar{x})^2/(n-1)$
- Here $\Sigma x=354$ and $n=15$, so $\bar{x}=354/15=23.600$ minutes. The sample variance is $s^2=11.400$ minutes^2 and sample SD is $s=3.376$ minutes.

Estimate	Value	Meaning
Population mean (estimated by $\bar{x}$)	23.600 min	Estimated average delivery time
Population variance (estimated by s^2)	11.400 min^2	Estimated variability in delivery times
Sample standard deviation	3.376 min	Typical spread around the sample mean

Interpretation: Delivery time is estimated at about 23.6 minutes on average. The SD shows order times vary around this mean; managers can use the estimate for service targets but should collect a larger sample across days and locations.

Problem 6 – Customer Spending

The 12 customer spending amounts are treated as a sample.

- Sample mean: $\bar{x} = \Sigma x/n$
- Sample SD: $s = \sqrt{[\Sigma(x-\bar{x})^2/(n-1)]}$
- $\Sigma x=\$1750$; $n=12$; $\bar{x}=\$145.83$. Sample variance $s^2=258.33$ dollars^2, so sample SD $s=\$16.07$.

Estimate	Result
Average spending	\$145.83
Sample variance	258.33
Sample standard deviation	\$16.07

Problem 7 – Product Ratings

Ratings from 20 customers are used to estimate the population's average product rating.

- Sample mean: $\bar{x} = \Sigma x/n$
- Sample variance: $s^2 = \Sigma(x-\bar{x})^2/(n-1)$
- Standard error of mean: $SE(\bar{x}) = s/\sqrt{n}$
- For n=20, $\bar{x}$=4.150, s^2=0.5553, s=0.7452, and SE=0.7452/√20=0.1666.

Quantity	Value	Interpretation
Sample mean	4.150	Average observed rating
Sample variance	0.5553	Variation among ratings
Standard error	0.1666	Estimated sampling uncertainty of the mean

Summary: The mean is close to 4 out of 5, suggesting generally favourable ratings. SE describes how much sample means would tend to vary across repeated samples of 20 customers; it is not the spread of individual customer ratings.

Problem 8 – Exam Scores

The 25 exam scores are treated as a sample.

- Sample mean: $\bar{x} = \Sigma x/n$
- Sample SD: $s = \sqrt{[\Sigma(x-\bar{x})^2/(n-1)]}$
- Standard error: $SE = s/\sqrt{n}$
- The sum of scores is 1890; mean=75.600, sample variance=15.667, sample SD=3.958, and SE=3.958/√25=0.792.

Statistic	Result	Meaning
Mean	75.600	Average score
Standard deviation	3.958	Spread of individual scores
Standard error	0.792	Precision of estimated mean

Comparison: The SD is larger than the SE because SD describes individual-score variability, while SE describes uncertainty in the sample mean and shrinks as sample size increases.

Problem 9 – Manufacturing Weights

A sample of 50 packets has mean weight 498 g and standard deviation 12 g.

- 95% CI for μ: $\bar{x} \pm t(0.975, n-1) \times s/\sqrt{n}$
- Standard error = $12/\sqrt{50} = 1.697$ g. With df=49, t critical ≈ 2.010. CI = $498 \pm (2.010 \times 1.697) = (494.59, 501.41)$ g.

Item	Result
Point estimate	498 g
Margin of error	3.41 g
95% confidence interval	(494.59, 501.41) g
Target 500 g	Inside the interval; plausible under this sample

Interpretation: The interval includes 500 g, so the data do not rule out a population mean of 500 g at the 5% level. This does not prove the true mean is exactly 500 g.

Problem 10 – Customer Satisfaction

Branch A reports mean rating 4.3 with 95% CI (4.1, 4.5); Branch B reports mean 4.0 with 95% CI (3.8, 4.2).

Branch	Sample mean	95% CI
A	4.3	(4.1, 4.5)
B	4.0	(3.8, 4.2)

- **Step 1: Compare means.** Difference = 4.3−4.0 = 0.3 points in favour of A.
- **Step 2: Compare intervals.** The intervals overlap between 4.1 and 4.2.

- **Step 3: Conclusion.** Branch A performs better descriptively, but overlap alone cannot establish whether the difference is statistically significant. Non-overlap is a conservative visual guide, and overlap is not a formal test. Sample sizes and standard errors (or raw data) are needed for a direct confidence interval/test of the difference.

Problem 11 – Average Battery Life

The manufacturer claims $\mu=10$ hours. The sample has $n=36$, $\bar{x}=9.5$ hours, and $s=1.8$ hours.

- H_0: $\mu = 10$; H_1: $\mu \neq 10$ (two-tailed)
- $t = (\bar{x}-\mu_0)/(s/\sqrt{n})$
- $SE=1.8/\sqrt{36}=0.30$. Test statistic $t=(9.5-10)/0.30=-1.667$, $df=35$. Two-tailed p-value=0.1045.

Decision rule	Result
Significance level	$\alpha=0.05$
p-value comparison	$0.1045 > 0.05$
Decision	Fail to reject H_0
Conclusion	Insufficient evidence that mean battery life differs from 10 hours

Problem 12 – Marketing Campaign

Sales before and after the campaign are paired by observation position; the company expects an increase.

- Define d = after − before. H_0: $\mu_d=0$; H_1: $\mu_d>0$
- Paired $t = \bar{d}/(s_d/\sqrt{n})$
- Differences are 8, 8, 8, 7, 6, 10, 9, 8. Mean difference $\bar{d}=8.000$; SD of differences=1.195; SE=0.423. Thus $t=8.000/0.423=18.931$, $df=7$; one-sided p-value=0.000000.

Measure	Result
Mean sales before	51.75
Mean sales after	59.75
Average increase	8.00
p-value	0.000000
Decision at 0.05	Reject H_0; evidence supports an increase

Interpretation: The p-value is the probability, assuming no true mean increase and the model assumptions, of observing a t statistic at least as large as the one obtained. Since $p<0.05$, the increase is statistically significant for these paired observations.

Problem 13 – Call Centre Response Time

A sample of 100 calls has mean response time 4.8 minutes and SD 1.5 minutes.

- Point estimate: $\hat{\mu} = \bar{x} = 4.8$ minutes
- 95% CI: $\bar{x} \pm t(0.975, n-1) \times s/\sqrt{n}$
- SE=$1.5/\sqrt{100}=0.15$ minutes; t critical=1.984. CI=$4.8\pm(1.984\times0.15)=(4.502, 5.098)$ minutes.

Quantity	Result
Population mean estimate	4.8 minutes
Standard error	0.15 minutes
95% CI	(4.502, 5.098) minutes

Frequentist interpretation: If this sampling and interval procedure were repeated many times, about 95% of the resulting intervals would contain the fixed true population mean. It is not correct to say the fixed mean has a 95% probability of lying in this particular interval.

Problem 14 – Defective Products

Of 500 products sampled, 22 are defective.

- Estimated defect proportion: $\hat{p} = x/n = 22/500$
- Approximate 95% CI: $\hat{p} \pm 1.96\sqrt{[\hat{p}(1-\hat{p})/n]}$
- $\hat{p}=0.044 = 4.4\%$. $SE=\sqrt{[0.044\times0.956/500]}=0.00917$.
  $CI=0.044\pm1.96(0.00917)=(0.0260, 0.0620)$, or (2.60%, 6.20%).

Quantity	Result
Estimated defect rate	4.40%
95% approximate CI	2.60% to 6.20%

Note: In repeated sampling, 95% of intervals constructed by this method would cover the true fixed defect rate. This is a normal-approximation interval; a Wilson interval can perform better for proportions.

Problem 15 – Email Spam Detection

Use Bayes' theorem to estimate the probability that an email is spam given that it contains the word "Offer."

- P(Spam | Offer) = P(Offer | Spam)P(Spam) / P(Offer)
- P(Offer) = P(Offer | Spam)P(Spam) + P(Offer | not Spam)P(not Spam)
- $P(Offer)=0.80\times0.25 + 0.10\times0.75 = 0.275$. Therefore
  $P(Spam|Offer)=(0.80\times0.25)/0.275= 0.7273 = 72.73\%$.

Probability	Value
P(Spam)	0.25
P(Offer | Spam)	0.80
P(Offer | not Spam)	0.10
P(Spam | Offer)	0.7273 (72.73%)

Interpretation: An email with "Offer" has approximately 72.73% probability of being spam given the stated base rates. The posterior is higher than the prior because "Offer" is much more common in spam emails.

Problem 16 – Disease Diagnosis

Disease prevalence is 2%, test sensitivity is 95%, and specificity is 90%.

- P(Disease | +) = [Sensitivity × Prevalence] / [Sensitivity×Prevalence + (1−Specificity)×(1−Prevalence)]
- P(D|+)=(0.95×0.02)/[(0.95×0.02)+(0.10×0.98)] = 0.019/0.117 = 0.1624 = 16.24%.

Component	Probability
Prevalence	2%
Sensitivity	95%
False-positive rate	10%
Posterior probability after positive test	16.24%

Note: This is not 95% because sensitivity means P(positive | disease), whereas the question asks P(disease | positive). Since the disease is rare, false positives among the much larger non-diseased group reduce the posterior probability. Confirmatory clinical assessment may be required.

Problem 17 – Website Conversion Rate

Page A receives 250 purchases from 5,000 visitors; page B receives 312 purchases from 5,200 visitors.

- Conversion rate: $\hat{p}$ = purchases / visitors
- H_0: pB=pA; H_1: pB>pA
- Pooled $\hat{p}$ = (xA+xB)/(nA+nB); $z=(\hat{p}B-\hat{p}A)/\sqrt{[\hat{p}pool(1-\hat{p}pool)(1/nA+1/nB)]}$
- Page A: 250/5000= 0.050 = 5.00%. Page B: 312/5200=0.060=6.00%. Pooled proportion=0.05510; SE under H_0=0.00452; z=2.213; one-sided p=0.01346.

Measure	Page A	Page B
Visitors	5000	5200
Purchases	250	312
Conversion rate	5.00%	6.00%

Conclusion: Since $p=0.01346<0.05$, reject H_0. Page B performs significantly better in conversion in this test. Its absolute improvement is 1.00 percentage points. The conclusion assumes comparable groups and valid visitor/purchase tracking.

Problem 18 – Mobile App Feature Test

Feature A: n=400, mean=18 min, SD=5; Feature B: n=420, mean=20 min, SD=6. Test whether B increases engagement.

- H_0: $\mu B - \mu A = 0$; H_1: $\mu B - \mu A > 0$
- Welch $t = (\bar{x}B - \bar{x}A) / \sqrt{(sA^2/nA + sB^2/nB)}$
- Difference = 20−18=2 minutes. $SE=\sqrt{(25/400+36/420)}=0.3850$. $t=2/0.3850=5.195$; Welch df≈803.9; one-sided p=1.29846e-07.

Item	Result
Mean A	18 minutes
Mean B	20 minutes
Mean difference	2 minutes
Welch t statistic	5.195
One-sided p-value	1.29846e-07
Decision	Reject H_0 at 0.05; evidence supports increased engagement

Conclusion: Feature B has higher average engagement and the difference is statistically significant. Business decisions should also consider retention, user satisfaction, and development costs.

Problem 19 – Recommendation System

CTR increases from 4.8% to 5.3%, with 100,000 users in each group.

- H_0: pNew=pOld; H_1: pNew>pOld
- $z=(\hat{p}New-\hat{p}Old)/\sqrt{[\hat{p}pool(1-\hat{p}pool)(1/nNew+1/nOld)]}$

- Old clicks ≈ 4,800; new clicks ≈ 5,300. Pooled CTR=0.05050; SE=0.000979; z=5.106; one-sided p=1.65e-07 (two-sided p≈3.29e-07).

Measure	Result
Old CTR	4.8%
New CTR	5.3%
Absolute gain	0.5 percentage points
Relative increase	10.42%
z statistic	5.106
One-sided p-value	1.65e-07

Conclusion: The difference is statistically significant at α=0.05. Practical significance asks whether the gain matters commercially: a 0.5 percentage-point gain represents about 500 extra clicks per 100,000 users. Evaluate downstream purchases, revenue, user experience and costs before concluding the algorithm is beneficial.

Problem 20 – Salary Prediction Model

Model A MAE=4200 and Model B MAE=4000. Across 50 paired test folds, mean difference is 200 and SD of differences is 350. Define d=MAE(A)−MAE(B), so positive values favour B.

- H_0: $\mu d=0$; H_1: $\mu d\neq 0$ (two-tailed paired t-test)
- $t = \bar{d}/(sd/\sqrt{n})$; 95% CI = $\bar{d} \pm t(0.975,n-1)\times sd/\sqrt{n}$
- SE=350/√50=49.497. t=200/49.497=4.041, df=49; two-sided p=0.00019. 95% CI for MAE(A)−MAE(B) = 200 ± 2.010×49.497 = (100.53, 299.47).

Test output	Result
Mean MAE: Model A	4200
Mean MAE: Model B	4000
Mean paired difference (A−B)	200
Paired t statistic	4.041
Two-sided p-value	0.00019
95% CI for difference	(100.53, 299.47)

Decision	Reject H_0 at 0.05; improvement is statistically significant

Conclusion: Model B has lower average absolute error, and the paired test indicates that the mean reduction is statistically significant. The positive confidence interval for A−B supports an improvement. Statistical significance should be considered alongside the practical importance of reducing MAE by 200 salary units.